

Sample Knowledge Base A short multi-topic reference document for testing a Retrieval-Augmented Generation (RAG) pipeline. It covers five unrelated topics so retrieval quality can be evaluated -- a good system should pull the right section for the right question.

**Section 1: The Water Cycle** The water cycle describes how water moves continuously through the Earth's atmosphere, surface, and underground. It begins with evaporation, where heat from the sun turns liquid water in oceans, lakes, and rivers into water vapor. This vapor rises into the atmosphere and cools, a process called condensation, forming clouds. When water droplets in clouds become too heavy, they fall back to Earth as precipitation, in the form of rain, snow, or hail. Some of this water flows over land as runoff, eventually returning to rivers and oceans, while some seeps into the ground to become groundwater. Plants also contribute through transpiration, releasing water vapor from their leaves back into the atmosphere. This continuous cycle is essential for distributing fresh water across the planet and supports nearly all terrestrial and aquatic ecosystems.

**Section 2: The French Revolution** The French Revolution began in 1789 and fundamentally transformed France's political and social structure. It was driven by widespread frustration with the absolute monarchy, severe economic hardship, and stark inequality between the privileged nobility and clergy and the common population. The storming of the Bastille prison on July 14, 1789 became a symbolic turning point, marking open rebellion against royal authority. Over the following years, France abolished feudal privileges, adopted the Declaration of the Rights of Man and of the Citizen, and eventually executed King Louis XVI in 1793. The revolution proceeded through several turbulent phases, including the Reign of Terror, a period of extreme political violence led by radical factions. It ultimately gave rise to Napoleon Bonaparte, who seized power in 1799, ending the revolutionary period while carrying forward many of its legal and administrative reforms across Europe.

**Section 3: How Vaccines Work** Vaccines work by training the immune system to recognize and fight specific pathogens without causing the disease itself. Most vaccines introduce a harmless piece of a virus or bacterium, such as a weakened form, an inactivated form, or just a fragment like a protein, into the body. The immune system responds by producing antibodies and creating memory cells that remember this specific invader. If the person is later exposed to the real pathogen, their immune system can respond much faster and more effectively than it would have without prior exposure, often preventing illness entirely or significantly reducing its severity. Some vaccines require multiple doses, called boosters, to strengthen and extend this immune memory over time. This principle, generating immunity without suffering the actual disease, is the foundation of modern immunization programs worldwide.

**Section 4: Basics of Compound Interest** Compound interest is the process where interest earned on an investment or loan is added back to the principal amount, so that future interest is calculated on this larger total, not just the original sum. This creates a snowballing effect over time, where growth accelerates the longer money remains invested. For example, one thousand dollars invested at a five percent annual interest rate, compounded yearly, would earn fifty dollars in the first year, but in the second year, interest is calculated on one thousand and fifty dollars rather than the original one thousand, producing slightly more growth than the prior year. The frequency of compounding, whether annual, monthly, or daily, also affects total returns, with more frequent compounding producing modestly higher growth for the same stated interest rate. This mechanism is central to long-term investing, retirement savings, and also works against borrowers who carry debt over long periods.

**Section 5: What Makes a Good Chess Opening** A good chess opening generally focuses on three core principles: controlling the center, developing pieces efficiently, and ensuring king safety. Controlling central squares, particularly the

four squares in the board's middle, gives pieces greater mobility and influence over the rest of the board. Developing pieces means bringing knights and bishops into active positions early, rather than moving the same piece repeatedly or delaying development in favor of premature attacks. King safety is typically addressed through castling, a special move that tucks the king behind a wall of pawns while simultaneously activating a rook. Strong openings avoid unnecessary pawn moves, avoid bringing the queen out too early where it can be attacked and lose tempo, and aim to complete development before starting aggressive attacks. Deviating from these principles is possible at high levels of play, but usually requires precise, memorized preparation to avoid falling into a disadvantageous position.